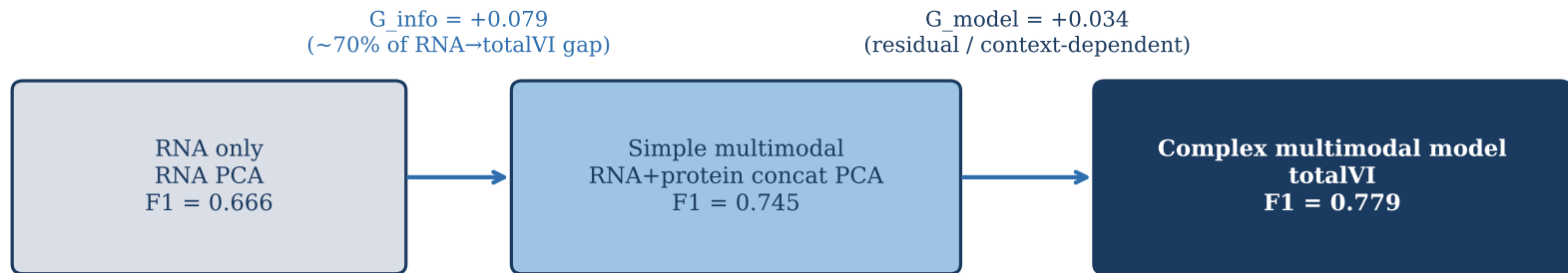


Figure 1. Central logic: information gain vs model-specific advantage



Stress-test the residual model-specific advantage under:
dataset transfer (inductive + target-data-access-matched) · protein degradation · training stochasticity · uncertainty diagnostics
External donor evidence (Lawlor): multimodal information value across 10 donors — not universal architecture superiority

Descriptive contrasts only (not causal variance components):
 $G_{\text{info}} = F1_{\text{concat}} - F1_{\text{RNA_PCA}}$ · $G_{\text{model}} = F1_{\text{totalVI}} - F1_{\text{concat}}$

Central question: how much multimodal gain is protein access, and when does a complex model add reliable value beyond simple multimodal features?